Detection of Bipolar Disorder from Behavioral and Psychological Characteristics Using Machine Learning

Manel Hattab
Department of Data Science
polytechnique de Sousse
Sousse, Tunisia
manel.hattab@polytechnicien.tn

Darine Mnasri
Department of Data Science
polytechnique de Sousse
Sousse, Tunisia
darine.mnasri@polytechnicien.tn

***Abstract*—Mental disorder diagnosis remains a complex clinical challenge, particularly for conditions such as bipolar disorder and major depressive disorder, where symptom overlap frequently leads to misclassification. This paper presents a machine learning pipeline for the automatic classification of four mental health conditions: Normal, Depression, Bipolar Type-1, and Bipolar Type-2. Using a structured clinical dataset of 120 patients with 17 behavioral and psychological features, we evaluate five classical classifiers through stratified 5-fold cross-validation and identify Random Forest as the best-performing model with 84.67% cross-validation accuracy and 87.5% on the unseen test set. We extend the study to two deep neural network architectures trained on a moderate SMOTE-augmented dataset of approximately 140 samples (targeting $\sim$35 examples per class). A rigorous three-way train/validation/test split (60/20/20) ensures that early stopping is guided by a fully independent real validation set, keeping the test set completely unseen until final evaluation. Experimental results confirm the effectiveness of model-specific and proportionate augmentation strategies on small clinical datasets.**

***Index Terms*—mental disorder classification, bipolar disorder, random forest, deep neural network, SMOTE, early stopping, machine learning, psychiatric diagnosis, behavioral features**

I. Introduction

This section describes the clinical context of the mental disorder diagnosis, existing classical and AI-based solutions and the proposed approach and structure of the paper.

A. General Context and Problem Statement

Mental health disorders represent one of the leading causes of disability worldwide. According to the World Health Organization, approximately one in eight people globally lives with a mental disorder [1]. Among these, mood disorders — particularly bipolar disorder and major depressive disorder (MDD) — are among the most frequently misdiagnosed conditions in clinical practice. The difference between Bipolar Type 1 Bipolar Type 2 Major Depressive Disorder and normal mental states is really hard to figure out.

Bipolar Type 1 and Bipolar Type 2 have some of the symptoms as Major Depressive Disorder, like mood swings, trouble sleeping and problems with thinking.These symptoms can be very different from one person to another.Doctors usually try to diagnose these conditions by talking to patients using rating scales like the Hamilton Depression Rating Scale and the Young Mania Rating Scale and watching patients over time.

This process takes a time and is not always accurate. It also depends a lot, on how experience the doctor has. Sometimes doctors misdiagnose Bipolar Type 1 and Bipolar Type 2. This can happen about 40 to 70 percent of the time. When this happens patients might not get the treatment and their condition can get worse. Bipolar Type 1 and Bipolar Type 2 are conditions that need to be diagnosed correctly so patients can get the help they need. [2].

B. Existing Solutions

Classical ways to diagnose problems include talking to patients in a structured way having them fill out questionnaires and using special tests to measure their thoughts and feelings. These tools have been. Are used a lot but they are not very good for looking at a lot of people at the same time and different people may get different results when they use them.

In the field of intelligence people have used machine learning to try to figure out what is going on with someones mental health. They have used different kinds of machine learning like Support Vector Machines, Logistic Regression and Random Forests to look at the information, from psychiatric diagnosis and they have gotten pretty good results. Mental health classification is what they are trying to do with these machine learning tools and psychiatric diagnosis is a part of that. [3], [4]. Deep learning has been explored for neuroimaging-based detection [5], while recent work leverages wearable sensor data to predict mood episodes [6]. Multimodal fusion approaches combining clinical data and brain imaging have achieved promising AUC scores above 0.85 [7], [9]. Despite this progress, most high-performing systems rely on private, large-scale datasets, and methods for handling small clinical datasets remain underexplored.

C. Proposed Approach

This work addresses the challenge of multi-class mental disorder classification on a small, publicly available clinical dataset. Our pipeline follows the CRISP-DM methodology, combining five classical classifiers benchmarked under stratified cross-validation with two feed-forward neural networks trained on moderately augmented data. The key methodological contribution is a strict three-way data split (60/20/20) combined with a targeted SMOTE strategy that produces approximately 140 augmented training samples — significantly smaller than common large-scale augmentation approaches — to better preserve the statistical properties of the original dataset while enabling stable neural network convergence.

D. Paper Structure

The remainder of this paper is organized as follows. Section II reviews related works. Section III describes the methodology in detail. Section IV presents experimental results. Section V discusses results and compares with related works. Section VI concludes with limitations and perspectives.

II. Related Works

In this section we review the most relevant machine learning approaches to bipolar disorder and mental disorder classification, and locate our contribution in the existing literature.

A growing body of literature has applied machine learning to the detection and classification of mood disorders. Table I summarizes the most relevant works reviewed in this study.

Birner et al. [3] applied Logistic Regression, Random Forest, and SVM to clinical and cognitive data, achieving an AUC of approximately 0.84. Their work highlights the value of cognitive assessments (e.g., Stroop test, d2-R) as discriminative features for bipolar classification. However, the use of a private dataset limits reproducibility.

A study published in *Neural Computing & Applications* [4] benchmarked SVM, RF, KNN, and ANN on biological biomarkers and clinical symptom scores, with SVM achieving the best accuracy in the 80–90% range, confirming that classical ML models remain competitive when features are well-engineered from domain knowledge.

BMC Psychiatry [5] proposed an SVM classifier using multimodal neuroimaging features (sMRI + fMRI), reaching 85–90% accuracy. While neuroimaging provides rich biological signals, such data are costly to acquire and unavailable in most clinical settings.

JMIR Medical Informatics [6] explored wearable sensor data (activity, sleep, heart rate) with multiple classifiers, achieving 83% for depression detection and 91% for mania. Tomasik et al. [8] combined online questionnaires with blood biomarkers using XGBoost, obtaining an AUROC of approximately 0.92 for BD vs. MDD differentiation.

Yang et al. [9] fused sMRI, fMRI, and DWI modalities with a majority vote ensemble, reaching 89.5% accuracy on 108 bipolar patients. Pigoni et al. [7] demonstrated that stacking SVM on multimodal clinical and imaging data could achieve an AUC of 0.88 for suicide attempt prediction in bipolar disorder.

A large-scale meta-analysis by Colombo et al. [10] covering 65 studies and over 11,000 participants reported a combined accuracy of 0.77 (CI: 0.75–0.80) across LDA, SVM, and RF classifiers applied to neuroimaging, biomarkers, and cognitive data.

In contrast to these works, our study focuses on a publicly available structured dataset without relying on neuroimaging or wearable devices. Our methodological contribution lies in a rigorous three-way split design and a moderate, model-specific augmentation strategy adapted to small clinical datasets.

Table I describes the landscape of machine learning approaches for mental disorder classification. Key studies demonstrate that classical models and deep learning achieve competitive results when properly applied to structured clinical data, though most rely on private datasets or expensive modalities like neuroimaging.

III. Methodology

In this section we describe the whole experimental pipeline including dataset description and preprocessing, model selection, data augmentation and the early stopping strategy for training neural networks.

A. Dataset Description

Source: The Mental Disorder Classification dataset is publicly available on Kaggle [11]. Expert clinicians collected it and it includes diagnostic records of 120 patients in 19 columns (including patient number and target label).

Dimensions: 120×17 input features (after preprocessing: Patient Number dropped, target separated).

Target classes:

- Normal (0): 30 patients-no diagnosed disorder
- Bipolar Type-1 (1): 28 patients-severe manic episodes
- Bipolar Type-2 (2): 31 patients-hypomania and depressive episodes
- Depression (3): 31 patients-Major Depressive Disorder

The dataset is approximately balanced across classes.

Feature types: The 17 features span three categories:

- *Binary (YES/NO):* Mood Swing, Suicidal Thoughts, Anorexia, Authority Respect, Try-Explanation, Aggressive Response, Ignore & Move-On, Nervous Breakdown, Admit Mistakes, Overthinking (10 features)
- *Ordinal:* Sadness, Euphoric, Exhausted, Sleep Disorder, encoded as {Seldom:1, Sometimes:2, Usually:3, Most-Often:4} (4 features)
- *Numeric score (1–10):* Sexual Activity, Concentration, Optimism (3 features)

Data quality: No missing values were detected. One anomaly was corrected: the "Suicidal Thoughts" column contained trailing whitespace values ("YES " instead of "YES"). No duplicate records were found.

TABLE I
SUMMARY OF RELATED WORKS ON MACHINE LEARNING FOR MENTAL DISORDER CLASSIFICATION

Reference	Year	Models	Contribution	Metrics	Best Result	Data / Limitations
Birner et al. [3]	2024	LR, RF, SVM	Clinical + cognitive data for early BD detection; ML comparison	AUC, F1-score	AUC $\approx$ 0.84	Private; cognitive scores (Stroop, d2-R); limited generalizability
Neural Comput. & Appl. [4]	2021	SVM, RF, KNN, ANN	BD classification from biomarkers + clinical scores; benchmarking	Accuracy, Sensitivity, Specificity	80–90%; SVM best	Private; biomarker profiles + symptom severity
BMC Psychiatry [5]	2020	SVM	Multimodal neuroimaging (sMRI + fMRI) for BD classification	Accuracy, AUC, Sensitivity	85–90%	Public + Private; MRI features; costly acquisition
JMIR Med. Inform. [6]	2025	LR, DT, KNN, RF, XGBoost	Wearable sensor data to predict manic/depressive episodes	Accuracy, F1, AUC	91% (mania)	Private; physiological data; requires wearable device
Tomasik et al. [8]	2021	XGBoost	BD vs. MDD from online questionnaire + blood biomarkers	AUROC, Accuracy	AUROC $\approx$ 0.92	Private; elevated mood and impulsivity as key features
Yang et al. [9]	2023	Fusion (sMRI, fMRI, DWI) + majority vote	Multimodal neuroimaging fusion for BD vs. controls	Accuracy	89.5%	Private; 108 BD-I + 78 controls; cerebellum region
Pigoni et al. [7]	2025	SVM + stacking	Suicide attempt prediction in BD; clinical + imaging fusion	AUC, Balanced Acc.	AUC 0.88	Private (163 patients); 56 clinical + cortical features
Colombo et al. [10]	2022	LDA, SVM, RF	Meta-analysis of ML for BD across 65 studies	Combined Accuracy	0.77 (CI 0.75–0.80)	11,336 participants; neuroimaging + biomarkers

B. Data Preprocessing

The preprocessing pipeline consists of five sequential steps applied before the train/validation/test split:

Step 1 – Binary encoding: YES/NO features are mapped to $\{1, 0\}$.

Step 2 – Ordinal encoding: Frequency-based ordinal features are mapped to integer ranks $\{1, 2, 3, 4\}$.

Step 3 – Score extraction: Scores expressed as "X From 10" are parsed to extract the integer numerator.

Step 4 – Target encoding: Diagnostic labels are encoded as integers $\{0, 1, 2, 3\}$.

Step 5 – Three-way stratified split: A single frozen split yields three independent subsets:

$$120 \xrightarrow{60/20/20} \underbrace{72}_{\text{Train}} + \underbrace{24}_{\text{Val}} + \underbrace{24}_{\text{Test}} \tag{1}$$

The split is applied once with `random_state=42` and kept fixed across all experiments.

TABLE II
ROLE OF EACH DATA SUBSET IN THE EXPERIMENTAL PIPELINE

Subset	Size	Role
Train	72	Weight updates (RF + NN)
Train (augmented)	$\sim$140	Weight updates (NN only)
Validation	24	Early stopping (NN only)
Test	24	Final evaluation only

Table II shows how the original 120 samples are partitioned into three independent subsets for a rigorous experimental design. The augmented training set is used only for neural network training.

For neural network training, `StandardScaler` is fitted exclusively on the augmented training set and applied (transform-only) to validation and test sets, preventing data leakage.

C. Classical Benchmark

Five classifiers are benchmarked on the 72 original training samples using `StratifiedKFold` with $k = 5$: Logistic Regression, Random Forest (200 estimators), SVM (RBF kernel), K-Nearest Neighbors ($k = 5$), and Decision Tree. The model achieving the highest mean cross-validation accuracy is selected for final test evaluation.

The Random Forest ensemble prediction is:

$$\hat{y} = \arg\max_{c} \sum_{t=1}^{T} \mathbf{1}[h_t(\mathbf{x}) = c] \tag{2}$$

where $T = 200$ trees, h_t is the t-th decision tree trained on a bootstrap sample, and $c \in \{0, 1, 2, 3\}$ represents the four diagnostic classes.

Random Forest is trained without augmentation. Preliminary experiments confirmed that applying SMOTE to the Random Forest training set degrades performance, because synthetic interpolations distort the feature distributions that the ensemble exploits internally through its bagging mechanism.

D. Moderate Data Augmentation for Neural Networks

With only 72 training samples, each epoch with a batch size of 32 produces approximately 2–3 mini-batches, which is insufficient for stable gradient-based learning. However, applying a $\times 10$ augmentation as in some prior works risks generating too many synthetic points relative to the dataset size, potentially distorting the original class distributions.

We therefore apply a **moderate two-stage augmentation pipeline** targeting approximately 35 samples per class ($\sim$140 total augmented samples):

Stage 1 – SMOTE [12]: Applied with `k_neighbors=3`, oversampling each class up to $n_{\text{target}} = \max(\text{class counts}) + 5$, with a minimum floor of 35 samples per class. This yields a balanced augmented training set of approximately 140 samples without overwhelming the dataset with synthetic data.

Stage 2 – Conservative noise injection: A single noise pass with `noise_level=0.05` is applied. Ordinal features are shifted by ± 1 within $[1, 4]$, binary features are flipped with probability 3%, and numeric scores are shifted by ± 1 within $[1, 10]$.

This moderate strategy deliberately limits augmentation to preserve the statistical coherence of the original clinical data, which is important when the base dataset is small and clinically curated.

E. Neural Network Architectures

The output layer of both networks applies the softmax function:

$$P(y = c \mid \mathbf{x}) = \frac{e^{z_c}}{\sum_{j=0}^{3} e^{z_j}} \tag{3}$$

where z_c is the pre-activation score for class c. The model is optimized by minimizing the categorical cross-entropy loss:

$$\mathcal{L} = -\sum_{i=1}^{N} \sum_{c=0}^{3} y_{ic} \log \hat{y}_{ic} \tag{4}$$

Simple Network: Three fully connected layers (64–32–4), ReLU activations, Dropout(0.3) then Dropout(0.2), Adam optimizer (lr $= 0.001$), softmax output. Trained for up to 150 epochs with early stopping patience of 20.

Improved Network: Two hidden layers (64–32) with BatchNormalization after each dense layer, Dropout(0.4) then Dropout(0.3), Adam optimizer (lr $= 0.001$), batch size of 16, softmax output. Trained for up to 150 epochs with early stopping patience of 20. This architecture is deliberately kept lighter than the simple network in terms of depth but heavier in regularization, reflecting the constraints of the $\sim$140-sample augmented dataset.

F. Early Stopping Strategy

Both networks use `EarlyStopping` monitoring `val_loss` with `restore_best_weights=True` and `ReduceLROnPlateau` (factor=0.5). The validation signal is provided exclusively by the independent real validation set $(X_{\text{val}}, y_{\text{val}})$:

```
model.fit(..., validation_data = (X_val, y_val))
```
(5)

This guarantees three properties: (1) early stopping is guided by real clinical data, not synthetic SMOTE samples; (2) the test set is never involved during training in any form; (3) reported test accuracy reflects genuine out-of-sample performance.

IV. Experimental Results

In this section, we report model performance under the same conditions, using the same set of test patients and metrics for each of the models evaluated. The first model is a classic benchmark and the second consists of 4 different neural networks. All models were tested on 24 patients not previously seen by the model.

A. Benchmark – Cross-Validation Results

Table III presents the 5-fold cross-validation results for all five classifiers on the 72 original training samples.

TABLE III
5-Fold Cross-Validation Results (Training Set, 72 Patients)

Model	Mean Accuracy	Std Dev
Random Forest	0.8467	0.0825
Logistic Regression	0.8076	0.0607
Decision Tree	0.7219	0.0625
SVM	0.6543	0.1220
KNN	0.6514	0.1042

Table III shows the cross-validation performance of all five classifiers. Random Forest dominates with 84.67% accuracy, while SVM and KNN perform significantly worse on this small dataset, indicating that ensemble methods are more robust than instance-based methods for this problem.

Random Forest achieved the highest mean cross-validation accuracy (84.67%) and was selected as the best classical model. Logistic Regression ranked second (80.76%), while SVM and KNN performed significantly lower on this small dataset (65.43% and 65.14% respectively).

TABLE IV
RANDOM FOREST CLASSIFICATION REPORT (TEST SET, 24 PATIENTS)

Class	Prec.	Recall	F1	Support
Normal	0.86	1.00	0.92	6
Bipolar Type-1	1.00	0.50	0.67	6
Bipolar Type-2	0.75	1.00	0.86	6
Depression	1.00	1.00	1.00	6
Macro Avg	0.90	0.88	0.86	24
Weighted Avg	0.90	0.88	0.86	24

B. Random Forest – Test Set Performance

The Random Forest trained on the 72 original samples achieves **87.5%** accuracy on the 24-patient unseen test set. Table IV reports the per-class metrics and Fig. 1 shows the confusion matrix.

Table IV presents the detailed per-class metrics for Random Forest on the test set. The model achieves perfect classification for Depression (F1 = 1.00) but struggles with Bipolar Type-1 (recall = 0.50), which is expected given the clinical similarity between mood disorders.

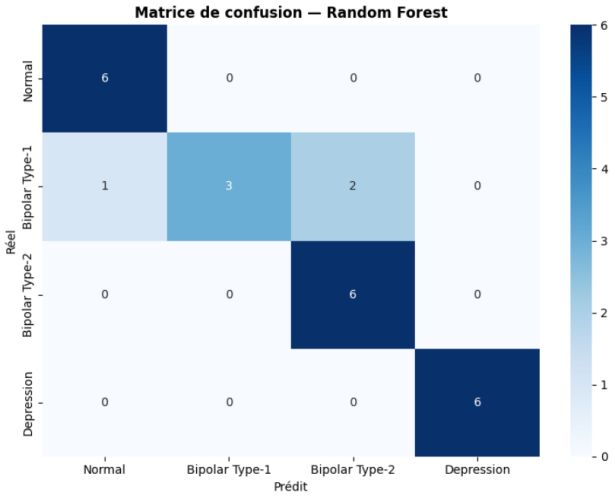


Fig. 1. Confusion matrix – Random Forest (test set, 24 patients).

Depression is classified with perfect precision and recall (F1 = 1.00). The main confusion occurs in Bipolar Type-1, where 3 out of 6 patients are misclassified, yielding the lowest recall (0.50) among all classes. This is consistent with the known clinical difficulty of distinguishing Bipolar Type-1 from other mood disorders based solely on behavioral features.

C. Neural Network Results

Table V compares the two neural network architectures on the same 24-patient test set. Both models were stopped early by the `EarlyStopping` callback before reaching their maximum epoch budget.

Table V compares the two neural network architectures trained on approximately 140 augmented samples. The Simple Network matches the Random Forest performance at 87.50%, while the Improved Network with additional regularization achieves only 83.33%, suggesting that overfitting is not the primary challenge on this small dataset.

TABLE V
NEURAL NETWORK RESULTS ON UNSEEN TEST SET (24 PATIENTS)

Model	Acc.	Macro F1	Loss	Max Epochs
Simple Network	87.50%	0.87	0.3903	150
Improved Network	83.33%	0.83	–	150

Tables VI and VII detail the per-class metrics of each network.

TABLE VI
SIMPLE NETWORK CLASSIFICATION REPORT (TEST SET, 24 PATIENTS)

Class	Prec.	Recall	F1	Support
Normal	0.86	1.00	0.92	6
Bipolar Type-1	1.00	0.67	0.80	6
Bipolar Type-2	0.75	1.00	0.86	6
Depression	1.00	0.83	0.91	6
Macro Avg	0.90	0.88	0.87	24

Table VI shows the per-class performance of the Simple Network. With a macro F1-score of 0.87, the network demonstrates stronger recall for Bipolar Type-1 (0.67) compared to Random Forest (0.50), though with a slight trade-off in Depression recall (0.83 vs 1.00).

TABLE VII
IMPROVED NETWORK CLASSIFICATION REPORT (TEST SET, 24 PATIENTS)

Class	Prec.	Recall	F1	Support
Normal	0.86	1.00	0.92	6
Bipolar Type-1	1.00	0.50	0.67	6
Bipolar Type-2	0.75	1.00	0.86	6
Depression	0.80	0.67	0.73	6
Macro Avg	0.85	0.79	0.80	24

Table VII presents the Improved Network's per-class metrics. The macro F1-score of 0.80 is notably lower than both the Simple Network (0.87) and Random Forest (0.86), particularly for Depression (F1 = 0.73), indicating that the increased regularization does not improve generalization on this dataset size.

V. DISCUSSION

This section discusses and compares the results obtained for all models, situates our results in the context of the related literature and discusses the main limitations of this study.

A. Comparative Summary Across All Models

Table VIII presents the final comparison across all models on the same unseen test set of 24 real patients.

Table VIII provides a comprehensive summary showing that Random Forest and Simple NN both achieve 87.50% accuracy through different approaches (bagging vs. augmentation + dropout). The Improved Network's lower performance demonstrates that model-specific design is crucial for small datasets.

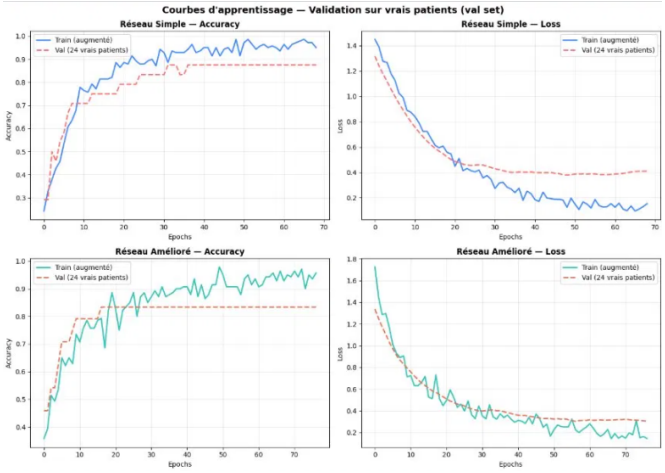


Fig. 2. Learning curves for both neural networks. Validation accuracy is computed on the independent 24-patient real validation set.

TABLE VIII
FINAL COMPARISON – UNSEEN TEST SET (24 REAL PATIENTS)

Model	Train Data	Anti-overfitting	Test Acc.
Random Forest	72 original	Bagging (200 trees)	**87.50%**
Simple NN	∼140 augmented	Dropout + Early Stopping	**87.50%**
Improved NN	∼140 augmented	Dropout + BatchNorm + ES	83.33%

B. Analysis of Results

Random Forest and the Simple Neural Network both achieve 87.5% accuracy on the unseen test set, confirming that two very different approaches — direct ensemble learning on original data versus gradient-based learning on moderately augmented data — can reach equivalent performance when properly designed.

The Improved Network (83.3%) underperforms relative to the Simple Network despite its added BatchNormalization layers. With only ∼140 augmented training samples, the combined effect of stronger regularization (Dropout 0.4/0.3 vs. 0.3/0.2) and batch normalization appears to slow convergence and prevent the network from reaching the same generalization level on the 24-patient test cohort. This illustrates that heavier regularization is not always beneficial at this dataset scale and that architecture choices must be calibrated to the actual training set size.

The moderate SMOTE strategy (∼140 samples instead of the 720 used in some prior experiments) proved sufficient for stable neural network convergence, while limiting the risk of distribution shift introduced by excessive synthetic interpolation. This validates the core design choice: augmentation volume should be proportionate to the original dataset size.

C. Comparison with Related Works

Our best result of **87.5% accuracy** on structured behavioral data compares favorably with the most relevant works in the literature on similar modalities:

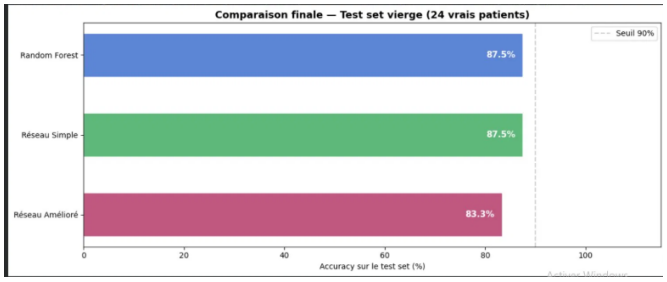


Fig. 3. Final accuracy comparison on the unseen test set (24 real patients). Random Forest and Simple NN both reach 87.5%, while the Improved NN achieves 83.3%.

- **Birner et al. [3]** achieved an AUC of approximately 0.84 using clinical and cognitive data (Stroop test, d2-R). While AUC and accuracy are not directly comparable, our result is consistent with this range on a comparable structured data format, obtained without private cognitive assessment tools.
- **Neural Comput. & Appl. [4]** reported 80–90% accuracy with SVM as the best model on biological biomarkers and clinical scores. Our Random Forest and Simple NN results (87.5%) fall within this range and demonstrate that similar performance is achievable on a fully public, smaller dataset using open methods.

Crucially, our 87.5% is obtained without neuroimaging (as in [5], [9]), without wearable sensors (as in [6]), and without private proprietary data, relying solely on 17 behavioral and psychological features from a public Kaggle dataset. This underscores the potential of structured behavioral indicators for low-cost, scalable diagnostic support in clinical settings.

D. Limitations

The primary limitation of this study is the small evaluation cohort (24 test patients). At this scale, a single misclassification shifts accuracy by 4.17%, making comparisons between models statistically fragile. Additionally, the dataset originates from Kaggle and has not undergone peer-reviewed clinical validation, which limits external generalizability. The synthetic nature of SMOTE samples introduces a distribution shift relative to real clinical populations that cannot be fully quantified without a larger independent cohort.

VI. CONCLUSION AND PERSPECTIVES

In this section, the main findings of this study are summarized, the major contributions are discussed and directions for future research are offered.

This paper presented a machine learning pipeline for the classification of four mental health conditions (Normal, Depression, Bipolar Type-1, Bipolar Type-2) using a structured clinical dataset of 120 patients and 17 behavioral and psychological features. A systematic benchmark of five classical classifiers under stratified 5-fold cross-validation identified Random Forest model stands out with an impressive (84.67% CV accuracy) as the best classical model, achieving 87.5% accuracy on the unseen test set. A Neural Network trained on

a moderate $\sim$140-sample SMOTE-augmented dataset matched this performance (87.5%), while a deeper Improved Network with stronger regularization reached 83.3%, confirming that additional regularization is not always beneficial at this dataset scale.

The main contribution of this work is two-fold. First, we empirically demonstrate that augmentation strategies must be both model and proportionate: Of the original 72 samples, Random Forest performed best, while neural networks required a moderate augmentation (140 samples) to converge stably without the risks of very large synthetic datasets. Second, we employ a rigorous 60/20/20 train/validation/test split, with early stopping based solely on the independent real validation set, so that the reported test accuracy is a fair estimate of generalization.

Future work should include: (1) the collection of a larger and clinically validated dataset to allow statistically robust model comparisons; (2) the exploration of feature selection methods to identify the most clinically predictive behavioral indicators; (3) the evaluation of gradient boosting approaches (e.g., XGBoost) which combine strong performance with interpretability; and (4) the investigation of multimodal approaches combining structured behavioral features with biological biomarkers or neuroimaging data, in the directions of recent literature [7], [9].

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